Chapter 1
Analytical Chemistry

Analytical Chemistry is the area of chemistry responsible for characterizing the composition of the matter both qualitatively and quantitatively along with its structure.

How is analytical chemistry different from chemical analysis ?

Analytical chemistry not only perform routine analysis on a routine sample like chemical analysis, it also works on

1. Improving established method
2. Extending existing method to new types of samples.
3. Developing new methods for measuring chemical phenomenon

For example,
Mining Engineers starts mining in a place, by comparing the cost of removing the ore with value of its contents. To estimate its value the analyse a sample of the ore. Providing the method for the analyze and also its improvements is the analytical chemist's responsibility. Once developed the routine, daily application of the method becomes the job of the chemical analyst.
Analytical chemists improve the ability of all chemists to make meaningful measurements on

1. Smaller samples
2. Complex samples
3. Shorter time scales
4. On species present at lower concentrations.

Analytical chemistry provides tools and methods necessary for research in the other four traditional area of chemistry (inorganic, organic, physical and bio-chemistry) as well as medicinal chemistry, toxicology, forensic chemistry, material science, geo-chemistry and environmental chemistry.

LANGUAGE OF ANALYTICAL CHEMISTRY

Analysis
A process that provides chemical or physical information about the constituents in the sample or the sample itself.
Analytes
The constituents of interest in a sample.
Matrix
All other constituents in a sample except for the analytes.
Or the remainder of the sample after analysis.
Determination
An analysis of a sample to find the identity, concentration, or properties of the analyte.
Measurement
An experimental determination of an analtye's chemical or physical properties.
Or to make the determination we measure one or more of the analyte's chemical or physical properties.

Classifying Analtical Technique

Analyzing a sample generates a chemical or physical signal whose magnitude is proportional to the amount of analyte in the sample. The signal may be anything we can measure; common examples are mass, volume and absorbance.

For our purpose it is convenient to divide analytical techniques into two general classes based on whether this signal is,

1. Proportional to an absolute amount of analyte.
2. Proportional to relative amount of analyte.

Consider two graduated cylinders, each containing 0.01M $Cu(NO_3)_2$. Cylinder 1 contains 10ml of Cu^{2+}. Cylinder 2 contains 20ml of Cu^{2+}.
If a technique responds to the absolute amount of analyte in the sample, then the signal due to the analyte S_A can be expressed as,

$$\mathbf{S_A = k\, n_A}$$

Where, n_A is the moles or grams of analyte in the sample, and k is a proportionality constant. Since cylinder 2 contains twice as many moles of Cu^{2+} as cylinder 1, analyzing the contents of cylinder 2 gives a signal that is twice that of cylinder 1.
A second class of analytical techniques are those that respond to the relative amount of analyte; thus,

$$\mathbf{S_A = k\, C_A}$$

Where, C_A is the concentration of analyte in the sample. Since the solution in the both the cylinders have the same concentration of Cu^{2+}, their analysis yields identical signals.

(1) Techniques responding to the absolute amount of analyte are called **Total analysis techniques.** Historically, most early analytical methods used total analysis techniques, hence they are often referred to as **'Classical techniques'**. Mass, volume and charge are the most common signals for total analysis techniques, and the corresponding techniques are gravimetry, titrimetry and coulometry.

(2) Techniques, such as spectroscopy, potentiometry and voltametry, in which the signal is proportional to the relative amount of analyte in a sample are called **Concentration techniques**. Since most concentration techniques rely on measuring an optical or electrical signal, they are also known as **Instrumental techniques**.

Selecting an analytical method

The requirements of the analysis determine the best method. In choosing a method, consideration is given to some or all the following design criteria: accuracy, precision, sensitivity, selectivity, robustness, ruggedness, scale of operation, analysis time, availability of equipment and cost.

(1) Accuracy

Accuracy is a measure of how closely the result of an experiment agrees with the expected result. It is a measure of the agreement between an experiment result and its expected value. Accuracy of a result is usually expressed in terms of error.

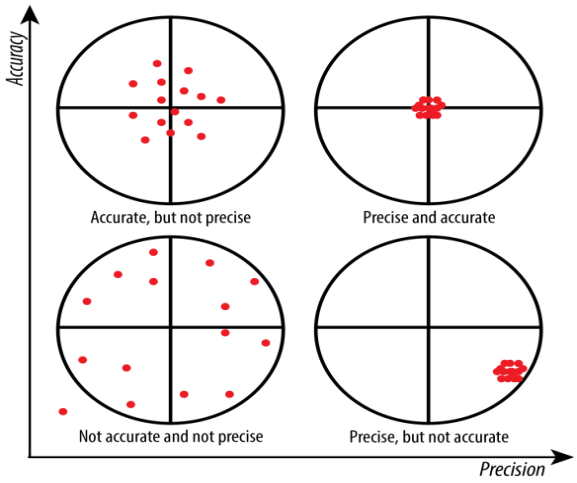
$$\text{Error} = \frac{\text{Experimental value} - \text{expected value}}{\text{expected value}} \times 100$$

Analytical methods can be divided into three groups based on the magnitude of their relative errors.

1. When an experimental result is within 1% of the correct result, the analytical method is highly accurate.
2. Methods resulting in relative errors between 1% and 5% are moderately accurate.
3. But methods of low accuracy produce relative errors greater than 5%

(2) Precision

Precision describes the reproducibility of measurements- in other words, the closeness of results that have been obtained in exactly the same way. The closer the agreement between individual analysis, the more precise the results.



Important points:

- It is important to realize that precision does not imply accuracy. That the data in figure (a) are more precise does not mean that the first set of results is more accurate. In fact, both sets of results may be very inaccurate.
- Like accuracy, precision depends on those factors affecting the relationship between the signal an and the analyte, i.e.,

$$S_A = k\, n_A \qquad (1)$$

$$S_A = k\, C_A \qquad (2)$$

- Precision also depends on the uncertainity in measuring the signal and the ease of handling samples reproducibility.
- In most cases the signal for a total analysis method can be measured with a higher precision than the corresponding signal for a concentration method.

(3) Sensitivity

Sensitivity is a measure of method's ability to distinguish between two samples that have different amount of analyte.

Sensitivity is the change in signal per unit change in the amount of analyte and is equivalent to the proportionality constant k, in equations (1) and (2)

$$S_A = k\, n_A \qquad \text{..... (1)}$$

$$S_A = k\, C_A \qquad \text{..... (2)}$$

The equations can rewritten as $k = \frac{\Delta SA}{\Delta nA}$ ..... (3) $k = \frac{\Delta SA}{\Delta CA}$ ..... (4)

(4) Selectivity

Selectivity is the ability of a method to determine the specific analyte in the presence of interferent contents.

An analytical method is selective if its signal is a function of only the amount of analyte present in the sample.

In the presence of an inteferent, equations (1) and (2) can be expanded to include a term corresponding to the interferent's contribution to the signal.

$$S_{Sample} = S_A + S_I = k_A n_A + k_i n_i \quad \text{.... (5) (total analysis method)}$$

$$S_{Sample} = S_A + S_I = k_A C_A + k_i C_i \quad \text{.... (6) (concentration method)}$$

Where, S_{samp} is the total signal due to presence f analyte en the interferent in the sample; k_A and k_i are the sensitivities for the analyte and the interferent, respectively: and n_i and C_i are the moles (or grams) and concentration of the interferent in the sample.

Selectively is a measure of a method's freedom from interferences as defined by the method's selectively coefficient.

Selectivity coefficient is a measure of a method's sensitivity for an interferent relative to the analyte (K_{AI}).

$$K_{AI} = k_i / k_A \quad \text{--(7)}$$

K_{AI} may be positive or negative depending on whether the interent's effect on the signal's opposite that of the analyte. A selectively coefficient greater than +1 or less than -1 indicates that the method is

Solving equation for k_i, in equation (7)

$$K_i = K_{A,I}{}^x k_A \quad \text{--(8)}$$

Substituting equation (8) in equations (5) and (6), we get

$$S_{Sample} = k_A(n_A + K_{A,I}{}^x n_i) \quad \text{(9)}$$

$$S_{Sample} = k_A(C_A + K_{A,I}{}^x C_i) \quad \text{(10)}$$

The selectively coefficient is easy to calculate if k_A and k_i can be independently determined. It is also possible to calculate $K_{A,I}$ by measuring S_{Sample} in the presence and absence of knon amounts of analyte and interferent.

Knowing the selectivity coeffiecient provides a useful way to evaluate an interferent's potential effect on an analysis. An interferent will not pose a problem as long as the term $K_{A,I}{}^x n_i$ in equation (9) is significantly smaller than n_A or $K_{A,I}{}^x C_i$ in equation (10) is significantly smaller than C_A.

LIMITATIONS OF ANALYTICAL METHODS

Errors: It is the numerical difference between true value/expected value and the experimental value. Error is an inverse measure of accuracy that is the smaller the error, the greater is the accuracy of an analysis.

How do we express error in a measurement?

There are two ways: 1) Absolute error

2) Relative error

Absolute error- It is a measure of quantity given by equation

$$E_a = X_i - X_t$$

Where Ea – absolute error

X_i – experimental value

X_t – true value

Relative Error- error expressed relative to the magnitude of the measured quantity

$$E_r = E_a/X_t$$

Where E_r – relative error

E_a – absolute error

X_t – true value

Percentage relative error = $(E_a/X_t)\times 100$

Numerical problems

1) In two separate determination the percentage iron in a haematite ore was found to be (a) 20.17 and (b) 19.80 taking the true value as 20. Calculate the absolute error as well as the relative error in percentage.

2) A student obtained a percent iso-octane in gasoline sample as 3.19% if the accepted value is 4% calculate the absolute error and relative errorin percentage.

Classification of errors:

I) Determinate errors/ Systematic errors- An error which can be described to a definite cause. It is unidirectional and can usually be determined and minimized.

3 types of systematic errors, based on sources

1) Instrumental errors- These are the errors arise from imperfections in measuring devices and instabilities in power supplies.

All measuring devices are the sources of systematic errors.
Reasons:

i *i. Use of defective instrument*

Example- Measuring devices such as pipettes, burettes, measuring cylinders, standard flasks etc. contain volumes that are slightly different from those indicated by their graduations.

i *ii. Environmental factors*

The use of glassware at a temperature which is significantly different from the temperature at which the glassware was calibrated. If a pipette calibrated at 200C is used at 300C, an error is introduced in measured volume.

i *iii. Power supply*

Instruments powered by electricity are very much proved to determinate errors because of:

a) Fluctuation in voltage of battery may introduce an instrumental error.

b) Increased resistance in circuits due to unclean electrical contacts.

c) Currents induced by 110/220V power lines affect electric instruments.

These errors can be easily detected and corrected.

2) Methodic errors- these arise from non-ideal behavior of physical/ chemical behaviour of reagents and reactions involved in a given analysis.

Example: in titrimetric analysis, we add small excess of titrant is usually added to make indicator function.
The non-ideality originates from

a) The slowness of reactions.

b) Incompleteness of reactions.

c) Non-specificity of reagents and occurrence of side reactions which interfere with the main process of measurement.

Since these errors are inherent in the method they cannot be easily detected and corrected.

3) Personal errors- these errors arise from erratic personal judgment and also from prejudice or bias. Many experimental measurements such as

a) The estimation of the position of a pointer between two scale divisions.

b) Judgment of the color of a solution at the end of titration.

c) Judgment of the level of a liquid with respect to a graduation on a burette or pipette are the sources of personal errors. Such errors vary from person to person and can be reduced to a minimum by experience and careful physical manipulation.

d) Prejudice or bias- most of us have a natural tendency to record the readings in such a way as to make the results fall close to a preconceived notion of the true value of the measurement. In order to secure accurate results we must get rid of bias at all cost.

How to minimize determinate errors

1) Every instrument used in analysis must be properly checked and calibrated.

2) Blanks should be used along with sample wherever possible.

3) Proper condition at which an instrument works are to be maintained properly

4) Particular measurement should be repeatedly made under identical condition and its constancy checked.

5) All possible personal errors must be predicted and eliminated in each measurement.

6) Personal errors are avoided (making arithmetic mistakes or reading scales wrongly must be strictly avoided).

7) The computing equipment must be checked for its correctness.

8) The reading taken must be re-read repeatedly and then recorded.

9) An experiment for obtaining data should not be performed when analyst is tired or not received complete training.

10) Standard method must be used.

II) Indeterminate errors/Random errors- A random unpredictable error which cannot be eliminated. These errors are the effect of many small uncontrollable variables and personal judgment that lead to uncertainty in a measured value.

Sources

- Measurement of any physical quantity requires making of the estimates. In an analysis, these can include
  - Judging the liquid level in a volumetric apparatus.
  - Estimating the position of a meter needle relative to the fixed scale.
- Any physical measurement can be affected by variables that are not controlled. These include variation in temperature, humidity or light intensity.

How to minimize indeterminate errors

Appropriate experimental design can reduce uncertainty to a tolerable minimum but there is no way it can be entirely eliminated. Hence all the physical measurements are subjected to a degree of uncertainty or independent error.

LIMIT OF DETECTION

Limit of detection is the lowest concentration of analyte in a sample that can be detected but not necessarily quantified.

Limit of detection can be determined

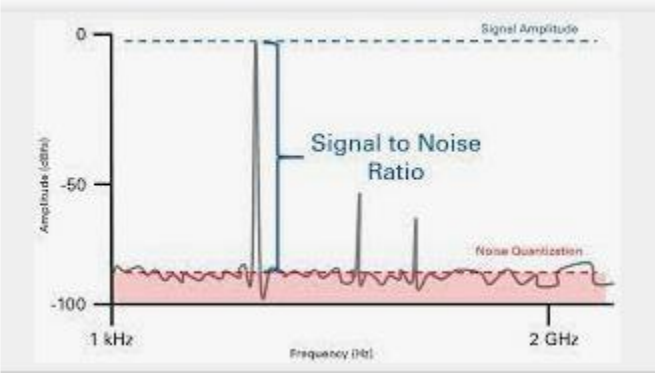
- • Visually
- • Signal to noise ratio
- • Standard deviation of the response and slope

Visual inspection

Non instrumental methods are based on the visual evaluation. The detection limit is determined by analysis of sample with known concentration of analyte and by establishing the minimum level at which the analyte can be readily detected.

Signal to noise ratio

This approach can only be applied to analytical procedures which exhibit baseline noise. Determination of the signal to noise ratio is performed by comparing measured signals from samples with known low concentration of the analyte with those of blank samples and establishing the minimum concentration at which analyte can be readily detected. A signal to noise ratio between 3:1 or 2:1 is generally acceptable for estimating LOD.



Standard deviation of response

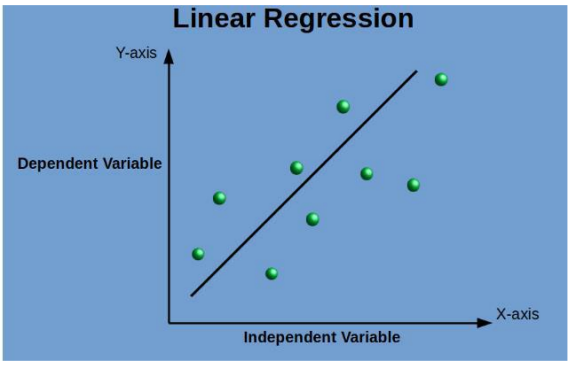
The detection limit may be expressed as
DL =
Where σ – standard deviation of response
S – sensitivity

Limit of quantification

Limit of detection is the lowest concentration of analyte in a sample that can be detected and also quantified.

External standard calibration:
Linear regression

So how do we set up a standard curve once the data have been collected ? First, a decision must be made regarding onto which axis to plot the paired sets of data. Traditionally, the concentration of the standards is represented on the x-axis and the observed readings on the y-axis. The x-axis data are called the independent variable and are assumed to be essentially free of error and y-axis data are called dependent variable and may have error associated with them. With modern day instruments, the error can be very small.



Once a linear relationship has been shown to have a high probability by the value of the correlation coefficient ***r,*** then the best straight line through the data points has to be estimated. This can often be

done by visual inspection of the calibration graph, but in many cases it is far more sensible to evaluate the best straight line by linear regression (the method of least squares).

The equation for the straight line is
y = bx + a
where y, the dependent variable, is plotted as a result of changing x, the independent variable.
To obtain the regression line 'y on x' , the slope b of the line and the intercept on the y-axis are given by the following equations:

$$b = \frac{[n(\sum xy) - (\sum x)(\sum y)]}{[n(\sum x^2) - (\sum x^2)]}$$

and **a = y-x**
where x is the mean of all values x1 and y is the mean of all values y1.

Correlation and regression

When using instrument methods it is often necessary to carry out a calibration procedure by using series of samples each having a known concentration of the analyte to be determined. A calibration curve is constructed by measuring the instrumental signal for each standard and plotting this response against concentration.

Correlation coefficient

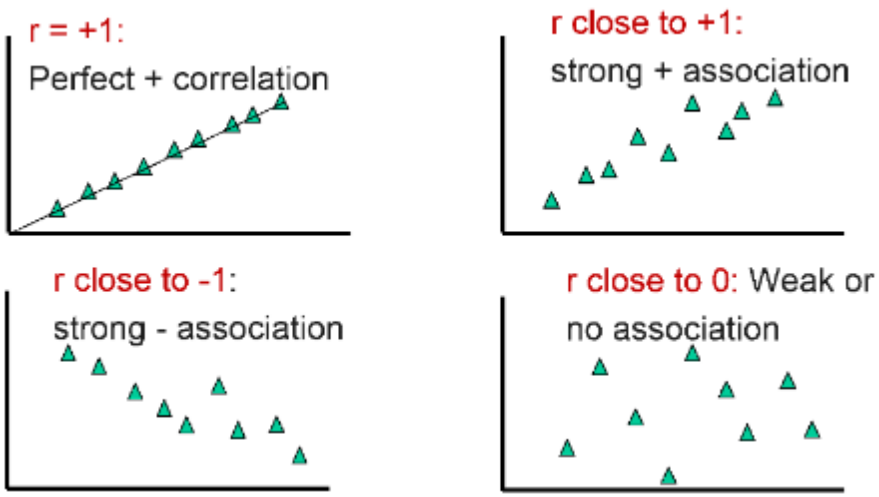
To establish whether there is a linear relationship between two variables x1 and y1, use pearsons correlation coefficient ***r*** :

Formula

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient
x_i = values of the x-variable in a sample
$\bar{x}$ = mean of the values of the x-variable
y_i = values of the y-variable in a sample
$\bar{y}$ = mean of the values of the y-variable

The sample correlation coefficient (r) is a measure of the closeness of association of the points in a scatter plot to a linear regression line based on those points. Possible values of the correlation coefficient range from -1 to +1, with -1 indicating a perfectly linear negative, i.e., inverse, correlation (sloping downward) and +1 indicating a perfectly linear positive correlation (sloping upward).



A correlation coefficient close to 0 suggests little, if any, correlation. The scatter plot suggests that measurement (y) do not change with increasing of x-axis values.

Linear dynamic range

Dynamic range - In the dynamic range, the response changes when the analyte concentration is changed but the relationship may be non-linear. If a response is linear, it can be specified as a dynamic linear range.

Working range - The working range is the range where the method gives results with an acceptable uncertainty. A working range can be wider than a linear range. This means that there is a correlation between the analyte concentration and the instrument's signal and concentrations of the analyte can be determined within this range - from LoQ to anomalies of the sensitivity.

Calibration range - The calibration range is the interval between the upper and the lower concentration of the analyte which can be determined with the demonstrated precision, accuracy and response function.

Linear range or linear dynamic Range – The range of concentrations where the signals are directly proportional to the concentration of the analyte in the sample.

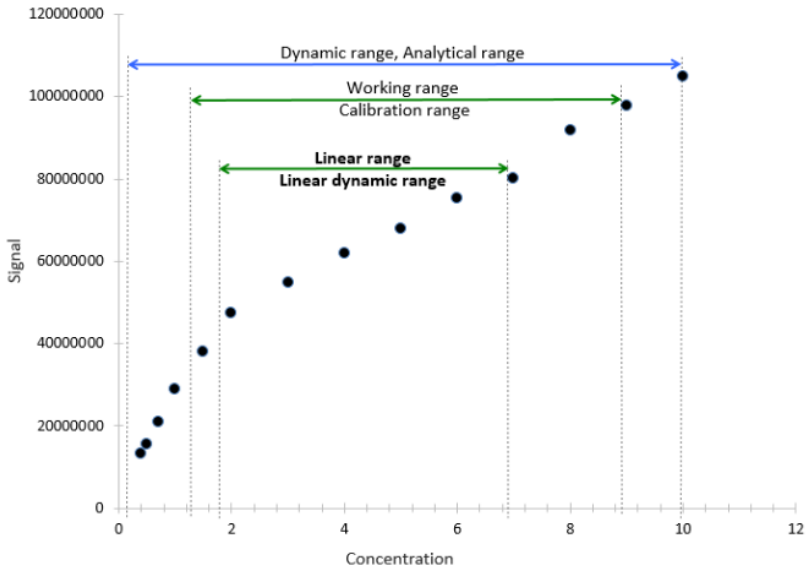


Figure 1. Different ranges